

# Yunfei Luo

Ph.D. Student at Halicioğlu School of Data Science and Computing, University of California San Diego

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## Research Interests

**General** AI/ML, Digital Health, Signal Processing, Multimodal Learning, Multitask Learning, Foundation Model, Reinforcement Learning

**Specific** Yunfei's research focuses on machine learning in the field of digital health, signal processing, multimodal modeling, and multitask learning. He aims to develop a robust and well-generalized methodology that enhances intelligent monitoring, supports timely interventions, and is deployable on edge devices.

**Applications** Mental Health, Substance Usage, Kidney Dialysis, Digital Health Monitoring

## Education

- 09.2023 - present **Ph.D. Student in Data Science, AI/ML/Digital-Health**  
GPA : 3.60/4.0 Halicioğlu School of Data Science and Computing (HSDSC), University of California (UC San Diego)  
➤ Advisor : 📧 Tauhidur Rahman  
➤ Research : Machine Learning, Multimodal Learning, Digital Health, Signal Processing
- 09.2021 - 05.2023 **Master of Science in Computer Science**  
GPA : 3.92/4.0 Manning College of Information and Computer Science (CICS), University of Massachusetts Amherst (UMass Amherst)  
➤ Advisor : Madalina Fiterau Brostean | 📧 Information Fusion Lab  
➤ Courses Taken : Advanced Machine Learning, Neural Networks, Probabilistic Graphical Models, Reinforcement Learning, Advanced Natural Language Processing, Advanced IoT
- 09.2017 - 05.2021 **Bachelor of Science in Computer Science | Bachelor of Science in Mathematics** (Dual Degrees)  
GPA : 3.84/4.0 CICS | College of Natural Sciences (CNS), UMass Amherst  
➤ Machine Learning and Computing Concentration

## Fellowships, Scholarships, Awards, and Honors

- 2023 - 2024 • Institute Fellowship | UC San Diego, HSDSC
- 2021 - 2023 • Bay State Fellowship | Full Tuition Scholarship, UMass Amherst, CICS
- 2017 - 2021 • Outstanding Undergrad Course Assistant Award, UMass Amherst, CICS  
• Chancellor's Award, UMass Amherst

## Teaching Experience

- 09.2021 to 06.2023 **Graduate Teaching Assistant, CICS, UMass Amherst**  
➤ CS 311 Introduction to Algorithm in Fall 2021 and Spring 2023  
➤ CS 590OP Applied Numerical Optimization, and CS 108 Foundation of Data Science in Fall 2022  
➤ CS 220 Programming Methodology in Spring 2022  
➤ Held office hours, led weekly discussion sessions, graded course works, answered questions on course forum, attended weekly TA meeting for discussing the progress/improvements of the course
- Javascript Algorithms Numerical Optimization Data Science Teaching

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| 01.2020<br>to 06.2021 | <b>Undergraduate Course Assistant, CICS, UMass Amherst</b> <ul style="list-style-type: none"> <li>&gt; CS 240 Reasoning under Uncertainty in Spring 2021, CS 311 Introduction to Algorithms in Fall 2020, CS 220 Programming Methodology in Spring 2020</li> <li>&gt; Helped with weekly discussion session, graded course works, and answered questions on Piazza</li> </ul> <div>Javascript Algorithms Probability Teaching</div> |
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## Work Experience

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| 06.2026<br>to present | <b>Data Science Research Intern, R&amp;D, Verily Health Inc.</b> <ul style="list-style-type: none"> <li>&gt; Research and Development in Machine Learning and Digital Health</li> </ul> <div>Research and Development Machine Learning Digital Health Signal Processing Foundation Model</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 07.2023<br>to 06.2026 | <b>Graduate Student Researcher, AI/ML/Digital-Health, HSDSC, UC San Diego</b> <ul style="list-style-type: none"> <li>&gt; Research focus on machine learning in signal processing and digital health. Detailed works are presented in the <i>Publications</i> section</li> </ul> <div>Research and Development Machine Learning Digital Health Signal Processing Natural Language Processing</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 06.2022<br>to 08.2022 | <b>Graduate Research Assistant, Center for Data Science, UMass Amherst</b> <ul style="list-style-type: none"> <li>&gt; Developed platform for Building Segmentation and Damage Assessment based on Satellite Imagery</li> <li>&gt; Integrated the Machine Learning pipeline along with User Interface for Imagery Fetching and Labeling</li> <li>&gt; Conducted Research and Experiments with various Finetuning setting and Machine Learning techniques to support selections of final-deployed hyper-parameters and models</li> <li>&gt; Delivered a presentation of the project to the Red Cross (funder and client), resulting in exceptional satisfaction and securing an additional year of funding for the initiative</li> <li>&gt; Worked as Internship in Data Science for Common Good program during the summer</li> </ul> <div>Computer Vision Machine Learning Disaster Response Software Development Full-Stack Development</div> |
| 06.2020<br>to 05.2021 | <b>Undergraduate Research Assistant, Institute for Applied Life Sciences, UMass Amherst</b> <ul style="list-style-type: none"> <li>&gt; Built pipelines for fMRI data preprocessing and statistical analysis</li> <li>&gt; Conducted Research and Experiments with Machine Learning models for Stroke Detection : Autoencoders, CNNs, and GANs</li> <li>&gt; Worked as Internship in Core Summer Intern program, and as Undergraduate Research Assistant after summer internship</li> </ul> <div>Machine Learning fMRI Data Preparation Data Engineering</div>                                                                                                                                                                                                                                                                                                                                                                                 |

## Publications

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| 2026<br>Published | <b>Toward Foundation Model for Multivariate Wearable Sensing of Physiological Signals, ACM Computing for Healthcare (ACM HEALTH), Journal</b><br>Yunfei Luo, Yuliang Chen, Asif Salekin, Tauhidur Rahman<br><div>Signal Processing Foundation Model Natural Language Processing Machine Learning Neural Networks Digital Health</div>                                                                                                                      |
| 2026<br>Published | <b>Personalized entropy-informed deep learning for identifying opioid misuse, Nature Mental Health, Journal</b><br>Yunfei Luo, Iman Deznabi, Bhanu Gullapalli, Mark Tuomenoksa, Eric Garland, Madalina Fiterau, Tauhidur Rahman<br><div>Signal Processing Natural Language Processing Heart Rate Variability Affective States Opioid Clinical Treatment Machine Learning Multimodal Fusion Digital Health</div>                                            |
| 2026<br>Published | <b>Automated Dysphagia Screening Using Noninvasive Neck Acoustic Sensing, IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Conference</b><br>Jade Chng, Rong Xing (Co-1st Author), Yunfei Luo (Co-1st Author), Kristen Linnemeyer-Risser, Tauhidur Rahman, Andrew Yousef, Philip A Weissbrod<br><div>Signal Processing Acoustic Signal Data Collection Clinical Treatment Applied Machine Learning Digital Health</div> |

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| 2025<br>Published | <b>Sounding out Dysphagia : Using a Digital Stethoscope to Assess Audiometric Differences in Swallowing, The Laryngoscope, Journal</b><br>Andrew Yousef, Sacha Moufarrej, David Zeng, Bao Luu, Rong Xing, James S. Crook, <b>Yunfei Luo</b> , Kristen Linnemeyer-Risser, Tauhidur Rahman, Philip A Weissbrod<br>Signal Processing Digital Health Dysphagia Acoustics Non-invasive |
| 2025<br>Published | <b>Opioid Misuse Detection from Cognitive and Physiological Data with Temporal Fusion Deep Learning, Drug and Alcohol Dependence, Journal</b><br>Bhanu Gullapalli, <b>Yunfei Luo</b> , Tauhidur Rahman, Eric L. Garland<br>Signal Processing Machine Learning Digital Health Addiction Psychophysiology Chronic Pain                                                              |
| 2024<br>Published | <b>Dynamic Clustering via Branched Deep Learning Enhances Personalization of Stress Prediction from Mobile Sensor Data, Nature Scientific Report, Journal</b><br><b>Yunfei Luo</b> , Iman Deznabi, Abhinav Shaw, Tauhidur Rahman, Madalina Fiterau Brostean<br>Multi-modal Fusion Methods Mental Health Machine Learning Neural Networks Time Series                              |
| 2024<br>Published | <b>Start Simple : Progressive Difficulty Multitask Learning, The Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL) 2024 SRW Proceedings, Conference</b><br><b>Yunfei Luo</b> , Yuyang Liu, Rukai Cai, Tauhidur Rahman<br>Natural Language Processing Multi-task Learning Machine Learning Neural Networks                                  |
| 2023<br>Published | <b>Agent Performing Autonomous Stock Trading under Good and Bad Situations, AI for Agent-Based Modelling (AI4ABM) in International Conference on Learning Representations (ICLR) 2023, Workshop</b><br><b>Yunfei Luo</b> , Zhangqi Duan<br>Reinforcement Learning Machine Learning Neural Networks                                                                                |

## Papers Under Preparation

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| 2026<br>Under Review   | <b>Toward World Modeling of Physiological Signals with Chaos-Theoretic Balancing and Latent Dynamics, -, Conference</b><br><b>Yunfei Luo</b> , Xi Chen, Yuliang Chen, Lanshuang Zhang, Md Mofijul Islam, Siwei Zhao, Peter Kotanko, Subhasis Dasgupta, Andrew Campbell, Rakesh Malhotra, Tauhidur Rahman<br>Signal Processing Foundation Model World Model Machine Learning Neural Networks Digital Health                                                                        |
| 2026<br>Preparing      | <b>Simulate before Acting : Counterfactual Exploration-Driven Reinforcement Learning for Acute Intervention in Intradialytic Hypotension, -, Journal</b><br><b>Yunfei Luo</b> , Siwei Zhao, Subhasis Dasgupta, Joseph M Mahaffy, Peter Kotanko, Rakesh Malhotra, Tauhidur Rahman<br>Signal Processing Reinforcement Learning Nephrology Clinical Treatment Machine Learning Digital Health                                                                                        |
| 2026<br>Await Decision | <b>Real Time Intradialytic Hypotension Forecasting Leveraging Modern Deep Learning Methods with Multimodal Fusion of Physiological Data and Electronic Health Records, npj Digital Medicine, Journal</b><br><b>Yunfei Luo</b> , Siwei Zhao, Subhasis Dasgupta, Joseph M Mahaffy, Peter Kotanko, Tauhidur Rahman (Equal Senior Author), Rakesh Malhotra (Equal Senior Author)<br>Signal Processing Nephrology Clinical Treatment Machine Learning Multimodal Fusion Digital Health |

## Patents

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| 2025 | US2025/016369 - System and Method for Personalized Closed-Loop Opioid Addiction Management with Mobile and Wearable Sensing of Administrations, Affective States, and Misuse Risk Scores ( <b>Full US Utility patent application filed on Feb 18, 2025</b> ) |
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## Voluntary Professional Activities

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### Talks

-  “Toward Foundation Model for Multivariate Wearable Sensing of Physiological Signals”, Lightning Talk at **Conference of Naval Applications of Machine Learning (NAML)**, March 3rd 2026
- “Foundation Model for Multivariate Wearable Sensing of Physiological Signals”, **Google Research Journal Club**, May 27th 2025
- “Health AI, foundation models, agents, and related research”, **Dexcom Journal Club**, April 18th 2023
-  “Integrating Machine Learning Models to Your Applications”, **Workshop in SackHacks V**, November 4th 2023

### Reviewer

- Neural Information Processing Systems (**NeurIPS**), 2026
  - Main Track 2026
- International Conference on Learning Representations (**ICLR**), 2026
  - Main Track 2026
  - Workshop on Time Series in the Age of Large Models, International Conference on Learning Representations (TSALM), 2026
  - Workshop on AI with Recursive Self-Improvement (RSI), 2026
- International Conference on Machine Learning (**ICML**), 2025 to 2026
  - Main Track 2026
  - Workshop on Foundation Models for Structured Data, International Conference on Machine Learning (FMSD), 2025 to 2026
- Conference on Computational Natural Language Learning (**CoNLL**), 2024 to 2026
- Proceedings of ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (**IMWUT**), 2024 to 2026

## Professional Certifications

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| 2022      | IBM Artificial Intelligence Engineering Professional Certificate |
| 2022      | Neuroscience and Neuroimaging Specialization Certificate         |
| 2020-2023 | TensorFlow Developer Professional Certificate                    |

## References

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### Tauhidur Rahman

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University of California San Diego  
Director of MOSAIC Lab  
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### Benjamin Smarr

Associate Professor of Data Science  
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### Rakesh Malhotra

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### Madalina Fiterau Brostean

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